Stochastic Fast Krasnosel'skiĭ–Mann Iteration with Unbiased Oracle Evaluations

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Abstract

We consider the problem of finding a fixed point of an averaged operator in a real Hilbert space using stochastic oracle evaluations. The operator and its fixed-point set are deterministic, whereas the oracle responses and the iterates are random. We focus on a stochastic implementation of the Fast Krasnosel'skiĭ–Mann method that uses one new oracle evaluation per iteration and retains the preceding evaluation. The objective is to identify noise conditions under which the last-iterate fixed-point residual decays at an accelerated almost-sure rate and the iterates converge weakly almost surely. This initial draft specifies the problem, the oracle model, and the iteration; stochastic convergence theorems and sharp noise thresholds remain to be established.

Key words. averaged operator, fixed-point problem, Fast KM, stochastic oracle, almost-sure convergence.

1 Introduction

We consider the problem of finding a fixed point of an averaged operator $T : \mathcal{H} \to \mathcal{H}$ in a real Hilbert space when only noisy evaluations of T are available. Many single-loop stochastic first-order methods are stochastic Krasnosel'skiĭ–Mann (KM) iterations of the form

$$x_{k+1} = (1 - \lambda_k)x_k + \lambda_k\big(Tx_k + e_k\big), \tag{1.1}$$

where e_k is the error of the kth oracle response [8, 9, 11]. Stochastic gradient descent, the stochastic proximal gradient method and stochastic backward–forward splitting are instances of this form; in the proximal gradient method, the error is in general conditionally biased.

For exact evaluations, the fixed-point residual of KM with a constant relaxation parameter satisfies $\|x_k - Tx_k\| = o(1/\sqrt{k})$ [4, Section 1]. The Fast KM method of Boţ and Nguyen [4, Algorithm 2.1] improves this to

$$\|x_k - Tx_k\| = o(1/k), \qquad x_k \rightharpoonup x^\star \in \operatorname{Fix} T, \tag{1.2}$$

with parameters that are fixed and do not vanish [4, Theorems 3.4–3.5]; see Section 3.2. Each update uses Tx_k and the value Tx_{k-1} computed in the preceding step. The forward–backward map above is covered by [4, Corollary 4.1].

With noisy evaluations, a single-query implementation stores the response $\widehat{T}_{k-1}(x_{k-1}) = Tx_{k-1} + e_{k-1}$ and uses it again in the next update. The error of the update then reads

$$\eta_k = \frac{s(k+\alpha/2)}{k+\alpha}\, e_k - \frac{sk}{k+\alpha}\, e_{k-1}, \qquad \mathbb{E}[\eta_k \mid \mathcal{F}_k] = -\frac{sk}{k+\alpha}\, e_{k-1}; \tag{1.3}$$

see Section 3.3. Two features distinguish this recursion from (1.1). First, the stored error e_{k-1} is known before the current query, so η_k is not a martingale difference. Second, the coefficient of the new error tends to s, so the parameters do not damp persistent noise. For $T = 0$ and $s = 1$, the iteration becomes $x_{k+1} = a_k x_k + a_k e_k - d_k e_{k-1}$ with $a_k = \frac{k+\alpha/2}{k+\alpha} \geq \frac{1}{2}$ and $d_k = \frac{k}{k+\alpha}$. If $\mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k] = \sigma^2 > 0$, conditional unbiasedness gives $\mathbb{E}\|x_{k+1} - Tx_{k+1}\|^2 \geq \sigma^2/4$ for every $k \geq 1$. Under bounded variance (N2), the residual therefore need not vanish. In contrast, stochastic KM tolerates bounded variance through vanishing relaxation parameters [8, Theorem 2.5].

These observations lead to the question studied in this paper:

For the single-query stochastic Fast KM method with fixed parameters, how fast must the variance σ_k^2 of the oracle errors decay for $k\|x_k - Tx_k\| \to 0$ and $x_k \rightharpoonup x_\infty \in \operatorname{Fix} T$ to hold almost surely along the same trajectory? Are the decaying-noise conditions (N6-ℓ^2), $\sum_k (k+1)^2\sigma_k^2 < \infty$, and (N6-$\ell^1$), $\sum_k (k+1)\sigma_k < \infty$, required by the algorithm or by the available proof technique?

With mini-batches of size $m_k \asymp k^q$ and $\sigma_k^2 = \sigma^2/m_k$, (N6-$\ell^2$) holds if and only if $q > 3$, and (N6-ℓ^1) if and only if $q > 4$ (Remark 2.3). The order $k^{3+\omega}$ is the batch growth used for related accelerated methods [14, Section 4.2 and Corollary 2].

Objectives. For the recursion of Section 3.3, under (N1) and finite second moments, we aim to establish:

1. almost-sure weak convergence of the iterates and vanishing residual under the variance budget $\sum_k \sigma_k^2 < \infty$, that is, $q > 1$, which is weaker than (N6-ℓ^1);
2. the almost-sure rate $k\|x_k - Tx_k\| \to 0$ under (N6-ℓ^2);
3. explicit examples that decide whether the weight $(k+1)^2$ in (N6-ℓ^2) can be reduced; for $T = 0$, the mean-square residual rate requires only $q > 2$, whereas $T = -\operatorname{Id}$ shows no cancellation of the stored error;
4. applications to stochastic gradient, proximal gradient and backward–forward methods, the proximal gradient method under the bias condition (N7).

The statements are research objectives of this draft; none of them is proved here.

Organization. Section 2 specifies the operator, the oracle model and the noise conditions. Section 3 recalls KM and Fast KM and states the stochastic recursion. Section 4 identifies stochastic gradient, proximal gradient and backward–forward methods as instances of the model.

1.1 Related work

Stochastic KM iterations. Bravo and Cominetti [8] study stochastic KM in finite-dimensional normed spaces. Their Theorem 2.5 gives almost-sure convergence of the iterates for martingale-difference noise with finite second moments, under summability conditions that hold for bounded variance and $\lambda_k = (k+1)^{-a}$ with $2/3 < a \leq 1$. Their Theorem 2.8 treats decaying, possibly biased errors with $\sum_k \lambda_k \mathbb{E}[\|e_k\| \mid \mathcal{F}_k] < \infty$ almost surely, allowing constant relaxation parameters; it yields convergence of the iterates without a rate. Combettes and Pesquet [9, Corollary 2.7] obtain the corresponding weak convergence in separable Hilbert spaces. Cortild, Cartis and Peypouquet [11] replace the uniform variance bound by finite variance at one fixed point and treat stochastic gradient and stochastic backward–forward splitting. In these results, the noise condition is balanced by the relaxation sequence; the Fast KM recursion has no such sequence.

Fast deterministic fixed-point and splitting methods. Fast KM [4] is extended by Boţ, Chenchene and Fadili [6] to overrelaxation and degenerate preconditioners, which include primal–dual splitting. Boţ, Nguyen and Zong [5, 7] obtain $o(1/k)$ rates for forward–backward and reflected forward–backward methods by working with the operator structure of the inclusion directly; these results are not consequences of Fast KM applied to an averaged map. The order $O(1/k)$ is worst-case optimal for nonexpansive operators [15, Theorem 4.6]. Yang et al. [19, Appendix A] identify Fast KM as a parameter choice of the inertial KM iteration and note that existing analyses concern exact evaluations or summable perturbations.

Stochastic accelerated methods. Almost-sure $o(1/k)$ residual rates are available for accelerated stochastic methods under three mechanisms.

- *Variance reduction.* VR-FKM [16] (finite sums), VFOSA [17, Theorem 14] and VAPEG [18, Corollary 3.4]. In VFOSA, the squared estimator error is summable with the weight $(k+1)^2$ almost surely, which is the weight in (N6-ℓ^2).

- *Growing mini-batches.* The AFP method [14, Corollary 2] uses delayed oracle values and batches of order $k^{3+\omega}$, $\omega > 0$.
- *Finite horizon or anchoring.* Yang et al. [19, Theorem 1] bound the root-mean-square last-iterate residual by $O(D/K + \sigma \log K/\sqrt{K})$ with one response per iteration and parameters depending on the horizon K. Anchored methods [1, 13] give expected last-iterate guarantees.

None of these results concerns the recursion of Section 3.3, which uses one fresh response per iteration, reuses the stored response and applies no variance reduction.

Complexity lower bounds. For stochastic oracles, $\Omega(\varepsilon^{-2})$ samples are necessary [12, Theorem 2]. Lee and Kim [13] show that the complexity $O(LD/\sqrt{\varepsilon} + \sigma^2/\varepsilon)$ is unattainable in general for monotone Lipschitz inclusions under their oracle assumptions. These bounds concern expected accuracy after a prescribed number of queries. They do not determine the noise decay required for an almost-sure rate along one trajectory of a fixed recursion, which is the question above.

1.2 Notation and preliminaries

Throughout, $\mathcal{H}$ is a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$, associated norm $\|\cdot\|$ and Borel sigma-algebra $\mathcal{B}$. Id denotes the identity operator on $\mathcal{H}$, and $\rightharpoonup$ and $\to$ denote weak and strong convergence in $\mathcal{H}$, respectively. An operator $R : \mathcal{H} \to \mathcal{H}$ is nonexpansive if $\|Rx - Ry\| \le \|x - y\|$ for all $x, y \in \mathcal{H}$. Following [4, Section 1], $T : \mathcal{H} \to \mathcal{H}$ is *θ-averaged*, $\theta \in (0, 1]$, if $T = (1 - \theta)\,\mathrm{Id} + \theta R$ for a nonexpansive R; the nonexpansive operators are exactly the 1-averaged ones. The set of fixed points of T is $\operatorname{Fix} T := \{x \in \mathcal{H} : Tx = x\}$. With $V := \mathrm{Id} - T$, the operator T is θ-averaged if and only if V is $\frac{1}{2\theta}$-cocoercive [4, equation (2.1)]:

$$\langle Vx - Vy, x - y \rangle \ge \frac{1}{2\theta} \|Vx - Vy\|^2, \qquad x, y \in \mathcal{H}. \tag{1.4}$$

For a proper, convex and lower semicontinuous $g : \mathcal{H} \to (-\infty, +\infty]$ and $\gamma > 0$,

$$\operatorname{prox}_{\gamma g}(x) := \operatorname*{argmin}_{y \in \mathcal{H}} \Big\{ g(y) + \frac{1}{2\gamma} \|y - x\|^2 \Big\}.$$

The underlying probability space is $(\Omega, \mathcal{F}, \mathbb{P})$. An $\mathcal{H}$-valued random variable is a measurable map $x : (\Omega, \mathcal{F}) \to (\mathcal{H}, \mathcal{B})$. Let $\mathscr{F} = (\mathcal{F}_k)_{k \ge 0}$ be a filtration of sub-sigma-algebras of $\mathcal{F}$. Following [9, Notation 2.1], we denote by $\ell_+(\mathscr{F})$ the set of sequences $(\rho_k)_{k \ge 0}$ of $[0, +\infty)$-valued random variables such that ρ_k is $\mathcal{F}_k$-measurable for every k, and set

$$\ell_+^p(\mathscr{F}) := \Big\{ (\rho_k)_{k \ge 0} \in \ell_+(\mathscr{F}) : \sum_{k \ge 0} \rho_k^p < +\infty \ \ \mathbb{P}\text{-a.s.} \Big\}, \qquad p \in (0, +\infty).$$

Equalities and inequalities between random variables hold $\mathbb{P}$-almost surely unless stated otherwise. For positive sequences, $a_k \asymp b_k$ means that a_k / b_k is bounded from above and bounded away from 0 [8, Section 1.2]; $O(\cdot)$ and $o(\cdot)$ refer to $k \to \infty$.

2 Problem formulation and stochastic oracle

2.1 Averaged fixed-point problem

Assumption 2.1 (Operator and solution set)**.** The deterministic operator $T : \mathcal{H} \to \mathcal{H}$ is θ-averaged for some $\theta \in (0, 1]$, and $\operatorname{Fix} T$ is nonempty.

The problem is

$$\text{find } x^\star \in \mathcal{H} \quad \text{such that} \quad Tx^\star = x^\star. \tag{2.1}$$

We measure accuracy by the fixed-point residual

$$r(x) := \|x - Tx\|, \qquad r_k := r(x_k). \tag{2.2}$$

Problem (2.1) is equivalently $Vx^\star = 0$, and V satisfies (1.4). The residual does not by itself provide a bound on the distance to $\operatorname{Fix} T$ without an additional error-bound condition.

2.2 Stochastic oracle and noise conditions

The sigma-algebra $\mathcal{F}_k$ of the filtration $\mathscr{F}$ contains all information available immediately before query k. At an $\mathcal{F}_k$-measurable point x_k, the oracle returns

$$\widehat{T}_k(x_k) = Tx_k + e_k, \qquad e_k = b_k + \xi_k, \tag{2.3}$$

where

$$b_k := \mathbb{E}[e_k \mid \mathcal{F}_k], \qquad \mathbb{E}[\xi_k \mid \mathcal{F}_k] = 0. \tag{2.4}$$

This is the oracle model of [19, Section 2.2, equation (6)]: b_k is $\mathcal{F}_k$-measurable, and ξ_k is $\mathcal{F}_{k+1}$-measurable and conditionally centered. Independence of errors across iterations is not assumed.

Assumption 2.2 (Finite second moments)**.** For every $k \geq 0$, $\mathbb{E}\|e_k\|^2 < \infty$.

Assumption 2.2 corresponds to the finite-variance part of (H1) in [8, Section 2.2]; it does not require the errors to decay. Set

$$\sigma_k^2 := \mathbb{E}[\|\xi_k\|^2 \mid \mathcal{F}_k]. \tag{2.5}$$

We use the following noise conditions; (N6) is the condition of [2, Theorem 3.1].

(N1) Conditional unbiasedness: $b_k = 0$ for all k.

(N2) Bounded variance: $\sigma_k^2 \leq \sigma^2$ almost surely for all k, for some constant $\sigma \geq 0$.

(N6) Decaying noise:

$$(s_k t_k \sigma_k)_{k\geq 0} \in \ell^2_+(\mathscr{F}) \qquad \text{or} \qquad (s_k t_k \sigma_k)_{k\geq 0} \in \ell^1_+(\mathscr{F}). \tag{2.6}$$

In (N6), s_k is the step size and $t_k := 1 + \sum_{i\geq k} \prod_{j=k}^{i} \alpha_j$ is generated by the extrapolation coefficients α_k. In [2, Theorem 3.1], the ℓ^2 form, condition (K_2), makes the noise term of the one-step energy inequality summable and yields the rate; the ℓ^1 form, condition (K_2^+), is used for the convergence of the iterates.

For the Fast KM recursion of Section 3, we take $s_k \equiv s$ and $\alpha_k = \frac{k}{k+\alpha}$, the factor shared by the two momentum terms of (3.2). Then $t_k = 1 + \alpha_k t_{k+1}$ gives $t_k = \frac{k+\alpha-1}{\alpha-1}$, and the two forms of (N6) read

$$\text{(N6-}\ell^2\text{)} \quad \sum_{k\geq 0} (k+1)^2 \sigma_k^2 < \infty, \qquad \text{(N6-}\ell^1\text{)} \quad \sum_{k\geq 0} (k+1)\sigma_k < \infty, \tag{2.7}$$

almost surely. This transfer of (N6) to Fast KM is a convention of the present paper. We use (N6) as a candidate sufficient condition; no convergence theorem for the recursion below is established in this draft.

For biased oracles, we add the following weighted summability of the bias, modeled on condition (H3) of [8, Theorem 2.8]:

(N7) Summable weighted bias: $(s_k t_k \|b_k\|)_{k\geq 0} \in \ell^1_+(\mathscr{F})$.

The bias is not averaged out by martingale cancellations, so (N7) is an ℓ^1 condition with the weights of (N6). For Fast KM, it reads $\sum_{k\geq 0}(k+1)\|b_k\| < \infty$ almost surely; whether this weight is needed is part of the question studied here. Condition (N7) arises for the stochastic proximal gradient method of Section 4.3.

Remark 2.3 (Noise conditions and mini-batches). Stochastic KM tolerates persistent noise through vanishing relaxation parameters: with $\lambda_k = (k+1)^{-a}$ and $2/3 < a \leq 1$, bounded variance suffices for the almost-sure convergence of the iterates [8, Theorem 2.5]. Condition (H3) of [8, Theorem 2.8] instead requires $\sum_k \lambda_k \mathbb{E}[\|e_k\| \mid \mathcal{F}_k] < \infty$; it admits bias and constant λ_k and gives convergence of the iterates without a rate. The Fast KM recursion has fixed parameters, so the noise itself must decay. If each response averages m_k conditionally independent samples of variance σ^2, then $\sigma_k^2 = \sigma^2/m_k$. For $m_k \asymp k^q$, (N6-ℓ^2) holds if and only if $q > 3$, and (N6-ℓ^1) holds if and only if $q > 4$.

Remark 2.4 (The target is deterministic). The operator T and the set $\operatorname{Fix} T$ are deterministic. The oracle responses and the iterates are random. A limit selected by the iteration may consequently be a random variable with values in $\operatorname{Fix} T$. The model does not require the sampled mappings to have a common fixed point or to be individually nonexpansive.

3 Fast KM iteration

3.1 The Krasnosel'skiĭ–Mann iteration

For the fixed-point problem (2.1), the Krasnosel'skiĭ–Mann (KM) iteration takes the form

$$x_{k+1} = (1-\lambda_k)x_k + \lambda_k T x_k, \qquad k \geq 0, \tag{3.1}$$

where $x_0 \in \mathcal{H}$ and $\lambda_k \in (0,1)$ is a relaxation parameter. Each update is a convex combination of the current iterate and its image under T. Under Assumption 2.1, for a fixed relaxation parameter $\lambda_k \equiv \lambda \in (0,1)$, the iterates converge weakly to a point in $\operatorname{Fix} T$ and the fixed-point residual satisfies $\|x_k - Tx_k\| = o(k^{-1/2})$; see [4, Section 1, equation (1.3)].

3.2 The Fast KM method

Boţ and Nguyen [4] accelerate the KM iteration using Nesterov's momentum updates. For fixed parameters $\alpha > 2$ and $0 < s \leq 1/\theta$, and initial points $x_0, x_1 \in \mathcal{H}$, their Fast KM method is

$$\begin{aligned} x_{k+1} = &\left(1 - \frac{s\alpha}{2(k+\alpha)}\right) x_k + \frac{(1-s)k}{k+\alpha}(x_k - x_{k-1}) + \frac{s\alpha}{2(k+\alpha)} T x_k \\ &+ \frac{sk}{k+\alpha}(Tx_k - Tx_{k-1}), \qquad k \geq 1. \end{aligned} \tag{3.2}$$

This is [4, Algorithm 2.1, equation (2.7)]. The update involves both the difference of successive iterates and the difference of successive operator values. Under Assumption 2.1, [4, Theorems 3.4–3.5] give

$$\|x_k - Tx_k\| = o(1/k), \qquad x_k \rightharpoonup x^\star \in \operatorname{Fix} T. \tag{3.3}$$

For $s = 1$, the coefficient of $x_k - x_{k-1}$ in (3.2) vanishes, and the iteration becomes

$$x_{k+1} = \left(1 - \frac{\alpha}{2(k+\alpha)}\right) x_k + \frac{\alpha}{2(k+\alpha)} T x_k + \frac{k}{k+\alpha}(Tx_k - Tx_{k-1}). \tag{3.4}$$

As observed in [4, Remark 2.1], this is a KM iteration with the additional extrapolation term $\frac{k}{k+\alpha}(Tx_k - Tx_{k-1})$. The extrapolation acts on the operator values. For $s \neq 1$, the explicit inertial term $\frac{(1-s)k}{k+\alpha}(x_k - x_{k-1})$ is also present.

3.3 Stochastic Fast KM

We replace the exact evaluations in (3.2) by the oracle responses (2.3) of Section 2.2. Choose deterministic $x_0, x_1 \in \mathcal{H}$ and fixed $\alpha > 2$, $0 < s \leq 1/\theta$. Initialize by computing and storing $\widehat{T}_0(x_0)$. For every $k \geq 1$, compute $\widehat{T}_k(x_k)$ and set

$$\begin{aligned} x_{k+1} = &\left(1 - \frac{s\alpha}{2(k+\alpha)}\right) x_k + \frac{(1-s)k}{k+\alpha}(x_k - x_{k-1}) + \frac{s\alpha}{2(k+\alpha)} \widehat{T}_k(x_k) \\ &+ \frac{sk}{k+\alpha}\big(\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1})\big). \end{aligned} \tag{3.5}$$

The preceding response $\widehat{T}_{k-1}(x_{k-1})$ is stored from the previous iteration. Hence, after initialization, each update requires one new oracle evaluation. The points x_{k-1} and x_k, together with the stored response, are $\mathcal{F}_k$-measurable; x_{k+1} is $\mathcal{F}_{k+1}$-measurable. The parameters are independent of a terminal horizon.

Substituting (2.3) into the difference of oracle responses yields

$$\widehat{T}_k(x_k) - \widehat{T}_{k-1}(x_{k-1}) = Tx_k - Tx_{k-1} + e_k - e_{k-1}. \tag{3.6}$$

Relative to the exact update (3.2) evaluated at the current iterates, the total additive error is

$$\eta_k = \frac{s\alpha}{2(k+\alpha)} e_k + \frac{sk}{k+\alpha}(e_k - e_{k-1}) = \frac{s(k+\alpha/2)}{k+\alpha} e_k - \frac{sk}{k+\alpha} e_{k-1}. \tag{3.7}$$

Since e_{k-1} is $\mathcal{F}_k$-measurable, (2.4) gives

$$\mathbb{E}[\eta_k \mid \mathcal{F}_k] = \frac{s(k+\alpha/2)}{k+\alpha}\, b_k - \frac{sk}{k+\alpha}\, e_{k-1}. \tag{3.8}$$

Under (N1), only the stored error remains in the conditional mean. These identities describe how the current and preceding oracle errors enter the stochastic iteration.

Our objective is to identify noise conditions under which

$$k\|x_k - Tx_k\| \longrightarrow 0 \qquad \text{almost surely,} \tag{3.9}$$

and, for a $\operatorname{Fix} T$-valued random variable x_∞,

$$x_k \rightharpoonup x_\infty \qquad \text{almost surely.} \tag{3.10}$$

4 Applications to stochastic optimization

This section identifies three single-loop stochastic methods for composite convex optimization as instances of the oracle model of Section 2.2. For each method we give the operator T, its averagedness constant, its fixed points, the oracle error and the noise conditions of Section 2.2 that the sampling scheme satisfies. The identification of stochastic gradient descent with stochastic KM follows [8, Section 4.3]; the splitting operators follow [4, Section 4] and [11, Section 3.2].

4.1 Setting

Consider

$$\min_{x\in\mathcal{H}}\ h(x) + g(x), \qquad h(x) = \mathbb{E}_\zeta[H(x,\zeta)]. \tag{4.1}$$

Assumption 4.1 (Composite problem)**.** The function $h : \mathcal{H} \to \mathbb{R}$ is convex and differentiable, and ∇h is L-Lipschitz continuous for some $L > 0$. The function $g : \mathcal{H} \to (-\infty, +\infty]$ is proper, convex and lower semicontinuous, and $\operatorname{argmin}(h+g) \neq \emptyset$.

By the Baillon–Haddad theorem [3], ∇h is $\frac{1}{L}$-cocoercive. A point $p \in \mathcal{H}$ minimizes $h+g$ if and only if $0 \in \partial g(p) + \nabla h(p)$.

Assumption 4.2 (Sampled gradients)**.** At an $\mathcal{F}_k$-measurable query point y_k, the gradient oracle returns an $\mathcal{F}_{k+1}$-measurable vector u_k such that

$$\mathbb{E}[u_k \mid \mathcal{F}_k] = \nabla h(y_k), \qquad \mathbb{E}\big[\|u_k - \nabla h(y_k)\|^2 \;\big|\; \mathcal{F}_k\big] \le \tau_k^2, \tag{4.2}$$

where $(\tau_k)_{k\ge 0} \in \ell_+(\mathscr{F})$.

The unbiasedness in (4.2) must be justified by the sampling model and, when u_k is built from $\nabla_x H$, by the interchange of differentiation and expectation. If u_k averages m_k samples $\nabla_x H(y_k, \zeta_{k,j})$, $j = 1, \dots, m_k$, that are conditionally independent given $\mathcal{F}_k$ and have conditional variance at most τ^2, then $\tau_k^2 \le \tau^2/m_k$.

The fixed-point residual of each operator below controls a first-order optimality measure of (4.1).

Lemma 4.3 (Residuals as optimality measures)**.** *Let Assumption 4.1 hold, let $\gamma > 0$, and let $x \in \mathcal{H}$.*

1. *If $p = \operatorname{prox}_{\gamma g}(x - \gamma\nabla h(x))$, then $\operatorname{dist}\big(0, \partial g(p) + \nabla h(p)\big) \le (\gamma^{-1} + L)\|x - p\|$.*
2. *If $p = \operatorname{prox}_{\gamma g}(x)$ and $T_{BF}x := p - \gamma\nabla h(p)$, then $\operatorname{dist}\big(0, \partial g(p) + \nabla h(p)\big) \le \gamma^{-1}\|x - T_{BF}x\|$.*

Proof. (i) The optimality condition of the proximal step gives $\gamma^{-1}(x-p) - \nabla h(x) \in \partial g(p)$. Adding $\nabla h(p)$ and using the Lipschitz continuity of ∇h yields the bound. (ii) The optimality condition gives $\gamma^{-1}(x-p) \in \partial g(p)$, and $\gamma^{-1}(x-p) + \nabla h(p) = \gamma^{-1}(x - T_{BF}x)$. □

4.2 Stochastic gradient descent

Let $g = 0$ and $0 < \gamma < 2/L$. Since $\mathrm{Id} - \frac{2}{L}\nabla h$ is nonexpansive [8, Section 4.3],

$$T_G := \mathrm{Id} - \gamma\nabla h = \Big(1 - \frac{\gamma L}{2}\Big)\,\mathrm{Id} + \frac{\gamma L}{2}\Big(\mathrm{Id} - \frac{2}{L}\nabla h\Big) \tag{4.3}$$

is $\frac{\gamma L}{2}$-averaged, and $\operatorname{Fix} T_G = \operatorname{argmin} h$. With the query point $y_k = x_k$, the response $\widehat{T}_k(x_k) = x_k - \gamma u_k$ has the error

$$e_k = -\gamma\big(u_k - \nabla h(x_k)\big), \qquad b_k = 0, \qquad \sigma_k \le \gamma\tau_k. \tag{4.4}$$

Hence (N1) holds; (N2) holds if $\sup_k \tau_k < \infty$; and (N6) holds if $(s_k t_k \tau_k)_{k\ge 0} \in \ell^2_+(\mathscr{F})$, respectively $\ell^1_+(\mathscr{F})$.

Substituting $\widehat{T}_k(x_k) = x_k - \gamma u_k$ into (3.5) and collecting terms, the stochastic Fast KM method applied to T_G reads

$$x_{k+1} = x_k + \frac{k}{k+\alpha}(x_k - x_{k-1}) - \frac{s\gamma(k+\alpha/2)}{k+\alpha}\,u_k + \frac{s\gamma k}{k+\alpha}\,u_{k-1}, \qquad 0 < s \le \frac{2}{\gamma L}. \tag{4.5}$$

Each iteration samples one gradient, at x_k; the vector u_{k-1} is the gradient sampled at x_{k-1} and stored. The update is a momentum step corrected by the difference of the current and stored sampled gradients. For exact gradients, [4, Theorems 3.4–3.5] give $\|\nabla h(x_k)\| = \gamma^{-1}\|x_k - T_G x_k\| = o(1/k)$ and weak convergence of x_k to a minimizer of h.

4.3 Stochastic proximal gradient

Let $0 < \gamma < 2/L$. The forward–backward operator

$$T_{FB} := \operatorname{prox}_{\gamma g} \circ (\mathrm{Id} - \gamma\nabla h) \tag{4.6}$$

is $\frac{2}{4-\gamma L}$-averaged and $\operatorname{Fix} T_{FB} = \operatorname{argmin}(h+g)$; see [4, Section 4] with $C = \nabla h$ and $B = 0$, or [10, Proposition 2.4]. With the query point $y_k = x_k$, the oracle returns

$$p_k := \widehat{T}_k(x_k) = \operatorname{prox}_{\gamma g}\big(x_k - \gamma u_k\big), \tag{4.7}$$

and the stochastic Fast KM method stores p_{k-1}:

$$x_{k+1} = \Big(1 - \frac{s\alpha}{2(k+\alpha)}\Big)x_k + \frac{(1-s)k}{k+\alpha}(x_k - x_{k-1}) + \frac{s\alpha}{2(k+\alpha)}\,p_k + \frac{sk}{k+\alpha}\,(p_k - p_{k-1}), \tag{4.8}$$

with $0 < s \le 2 - \frac{\gamma L}{2}$. Each iteration samples one gradient and evaluates one proximal map.

Lemma 4.4 (Error of the forward–backward oracle)**.** *Under Assumptions 4.1 and 4.2 with $y_k = x_k$, the error $e_k = p_k - T_{FB}x_k$ satisfies*

$$\|e_k\| \le \gamma\|u_k - \nabla h(x_k)\|, \qquad \|b_k\| \le \gamma\tau_k, \qquad \sigma_k \le \gamma\tau_k. \tag{4.9}$$

Proof. The first bound follows from the nonexpansiveness of $\operatorname{prox}_{\gamma g}$. By the conditional Jensen inequality, $\|b_k\| \le \mathbb{E}[\|e_k\| \mid \mathcal{F}_k] \le (\mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k])^{1/2} \le \gamma\tau_k$. Finally, $\sigma_k^2 = \mathbb{E}[\|e_k\|^2 \mid \mathcal{F}_k] - \|b_k\|^2 \le \gamma^2\tau_k^2$. □

Remark 4.5 (Unbiasedness of the full operator evaluation)*.* An unbiased gradient passed through a nonlinear proximal map need not produce an unbiased estimate of $T_{FB}x_k$. For example, let $P(v) = \max\{v, 0\}$ and let w be an integrable, nonzero symmetric scalar random variable. Then

$$\mathbb{E}[P(-\gamma w)] = \frac{\gamma}{2}\mathbb{E}|w| > 0 = P(0).$$

Thus (N1) fails in general, and the bias is controlled only through (4.9).

Lemma 4.4 shows that $(s_k t_k \tau_k)_{k\ge 0} \in \ell^1_+(\mathscr{F})$ implies (N7) and both forms of (N6). The bound $\|b_k\| \le \gamma\tau_k$ is of first order in τ_k, so the bias requires an ℓ^1 condition. With mini-batches $m_k \asymp k^q$ and $\tau_k^2 = \tau^2/m_k$, this means $q > 4$. By Lemma 4.3(i), the residual $\|x_k - T_{FB}x_k\|$ bounds $\operatorname{dist}(0, \partial g(T_{FB}x_k) + \nabla h(T_{FB}x_k))$ up to the factor $\gamma^{-1} + L$.

4.4 Stochastic backward–forward splitting

Exchanging the order of the two steps gives the backward–forward operator

$$T_{BF} := (\mathrm{Id} - \gamma\nabla h) \circ \mathrm{prox}_{\gamma g}, \qquad 0 < \gamma < 2/L, \tag{4.10}$$

which is also $\frac{2}{4-\gamma L}$-averaged, since the averagedness constant of a composition in [10, Proposition 2.4] is symmetric in the two factors. From Lemma 4.3(ii) and the converse computation,

$$\mathrm{Fix}\, T_{BF} = (\mathrm{Id} - \gamma\nabla h)\big(\mathrm{argmin}(h+g)\big), \qquad \mathrm{prox}_{\gamma g}(\mathrm{Fix}\, T_{BF}) = \mathrm{argmin}(h+g). \tag{4.11}$$

Indeed, if $p \in \mathrm{argmin}(h+g)$ and $x = p - \gamma\nabla h(p)$, then $\gamma^{-1}(x-p) = -\nabla h(p) \in \partial g(p)$, so $\mathrm{prox}_{\gamma g}(x) = p$ and $T_{BF}x = x$.

The proximal step is exact, so the query point $y_k = \mathrm{prox}_{\gamma g}(x_k)$ is $\mathcal{F}_k$-measurable, and the response $\widehat{T}_k(x_k) = y_k - \gamma u_k$ has the error

$$e_k = -\gamma\big(u_k - \nabla h(y_k)\big), \qquad b_k = 0, \qquad \sigma_k \le \gamma\tau_k. \tag{4.12}$$

Thus (N1) holds, and only (N6) is required of the sampling scheme. This ordering is the stochastic backward–forward splitting (SBF) of [11, Section 3.2 and Corollary 3.4] and the backward–forward reformulation used in [16]. The point of interest is the shadow $p_k := \mathrm{prox}_{\gamma g}(x_k)$, for which Lemma 4.3(ii) gives $\mathrm{dist}(0, \partial g(p_k) + \nabla h(p_k)) \le \gamma^{-1}\|x_k - T_{BF}x_k\|$. Weak convergence of x_k does not by itself give weak convergence of p_k, since $\mathrm{prox}_{\gamma g}$ is not weakly continuous in general.

4.5 Summary

Table 1 collects the three identifications. The batch exponents refer to $m_k \asymp k^q$ and $\tau_k^2 = \tau^2/m_k$.

Table 1: Stochastic first-order methods as instances of (2.3).

Method	T	θ	Bias b_k	Conditions; batch exponent
SGD	$\mathrm{Id} - \gamma\nabla h$	$\frac{\gamma L}{2}$	0	(N1), (N6); $q > 3$ for ℓ^2, $q > 4$ for ℓ^1
Prox. gradient	$\mathrm{prox}_{\gamma g} \circ (\mathrm{Id} - \gamma\nabla h)$	$\frac{2}{4-\gamma L}$	$\|b_k\| \le \gamma\tau_k$	(N6), (N7); $q > 4$
Backward–forward	$(\mathrm{Id} - \gamma\nabla h) \circ \mathrm{prox}_{\gamma g}$	$\frac{2}{4-\gamma L}$	0	(N1), (N6); $q > 3$ for ℓ^2, $q > 4$ for ℓ^1

The order of the proximal and gradient steps decides whether the oracle is biased. In the forward–backward order the bias bound is of first order in τ_k, which forces an ℓ^1 condition; the backward–forward order keeps (N1) at the price of measuring optimality at the shadow point p_k.

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